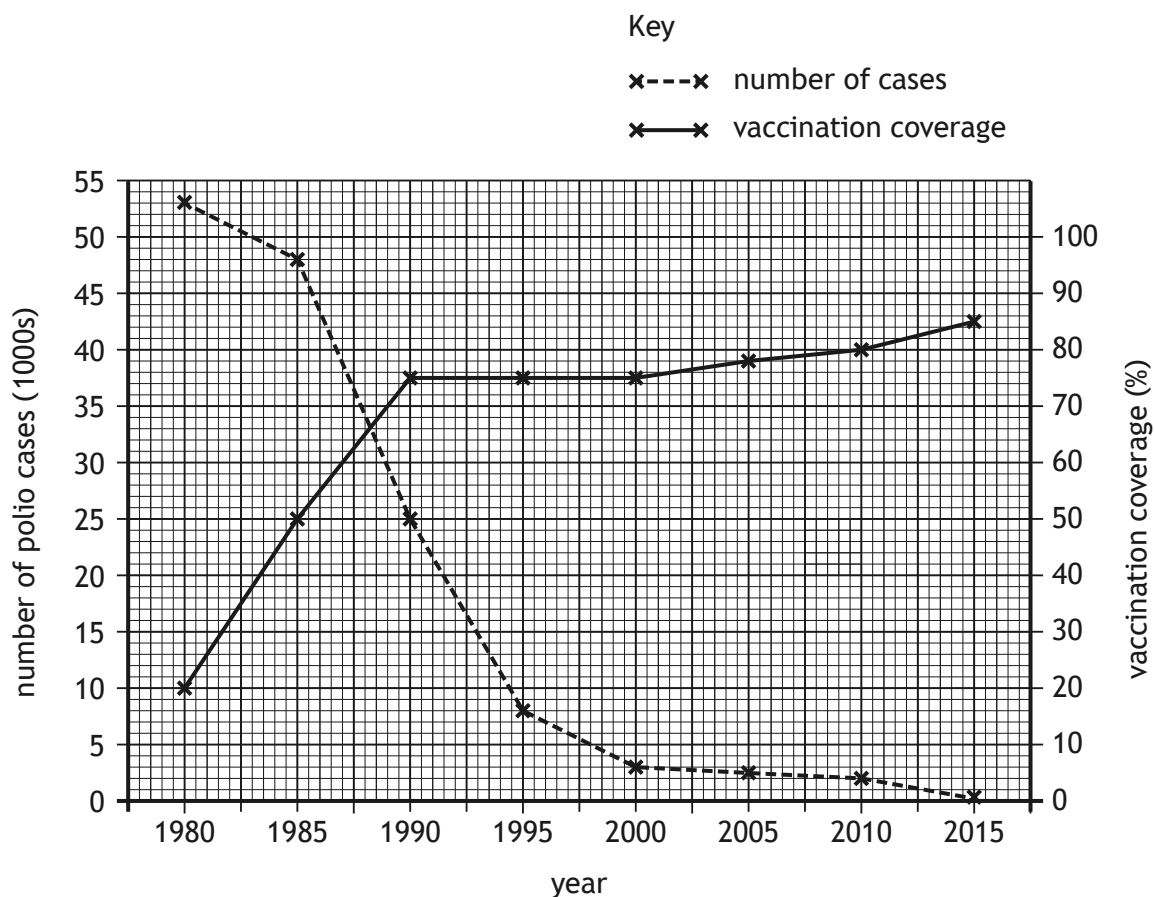


14. Polio is a viral disease that has been eradicated in most countries due to mass vaccination programmes.

The **graph** shows changes in the global number of polio cases and the percentage of the world population vaccinated between 1980 and 2015.



- (a) (i) State the vaccination coverage achieved when the number of polio cases was 8000.

1

_____ %

- (ii) Calculate the percentage decrease in polio cases in the 30-year period from 1980 to 2010.

1

Space for calculation

_____ %



* X 8 4 0 7 6 0 1 2 4 *

14. (a) (continued)

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- (iii) Use data from the graph to describe the changes in vaccination coverage between 1980 and 2000.

2

- (b) Polio is still present in some developing countries.

The **table** shows the number of polio cases in one of these countries between 2000 and 2010.

Year	2000	2002	2004	2006	2008	2010
Number of polio cases	90	50	20	15	10	15

- (i) Using data in the table and graph, calculate the simple whole number ratio of the global number of cases of polio to the number of cases in this country in 2000.

1

Space for calculation

_____ : _____
global country

- (ii) In order to establish herd immunity for polio, there has to be at least 85% vaccination coverage.

If the population of this country is 2 million people, calculate the number of individuals that must be vaccinated to establish herd immunity for polio in this country.

1

Space for calculation

- (iii) Suggest why the herd immunity threshold may not be reached in this developing country despite a 10-year mass vaccination programme.

1

